

EC4219: Software Engineering

Lecture 3 — First-Order Logic

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First-Order Logic (FOL)

- An extension of propositional logic (PL) with predicates, functions, and quantifiers.
- FOL is also called predicate logic, and the first-order predicate calculus.
- FOL is expressive enough to reason about programs.
- While the validity of PL formulas is decidable, the validity of FOL formulas is not.

Terms (Variables, Constants, and Functions)

- Terms are the objects that we are reasoning about.
- Terms in FOL evaluate to values other than truth values, such as integers, strings, or lists.
- Terms in FOL are defined by the grammar below:

$$t \rightarrow x \mid c \mid f(t_1, \dots, t_n)$$

- ▶ Basic terms are variables (denoted x) and constants (denoted c).
 - ▶ Composite terms are functions. When a function takes n terms as arguments, we say that the function is an n -ary function (or, the function has the arity n).
 - cf) A constant can be viewed as a 0-ary function.
- (Example) $g(x, b)$: a binary function g applied to a variable x and a constant b

Predicates

- The propositional variables of PL are generalized to predicates in FOL.
- An n -ary predicate takes n terms as arguments.
- A FOL propositional variable is a 0-ary predicate.
- For example, $p(f(x), g(x, f(x)))$ is a binary predicate applied to two terms.

- **Atom:** basic elements
 - ▶ truth symbols $\perp$ (“false”) and $\top$ (“true”)
 - ▶ n -ary predicates applied to n terms
- **Literal:** an atom α or its negation $\neg\alpha$.
- **Formula:** a literal, application of a logical connective to formulas, or the application of a quantifier to a formula.

F	$\rightarrow$	$\perp \mid \top \mid p(t_1, \dots, t_n)$	atom
		$\mid \neg F$	negation (“not”)
		$\mid F_1 \wedge F_2$	conjunction (“and”)
		$\mid F_1 \vee F_2$	disjunction (“or”)
		$\mid F_1 \rightarrow F_2$	implication (“implies”)
		$\mid F_1 \leftrightarrow F_2$	iff (“if and only if”)
		$\mid \exists x.F[x]$	existential quantification
		$\mid \forall x.F[x]$	universal quantification

Notations: Quantification

- In $\forall x.F[x]$ (or $\exists x.F[x]$), x is the quantified variable, and $F[x]$ is the **scope** of the quantifier $\forall x$ (or $\exists x$). We say x in $F[x]$ is **bound** (by the quantifiers).
- $\forall x.\forall y.F[x, y]$ can be abbreviated by $\forall x, y.F[x, y]$.
- By convention, the scope of the quantified variable extends as far as possible. For example, consider

$$\forall x. \overbrace{p(f(x), x) \rightarrow (\exists y. \underbrace{p(f(g(x, y)), g(x, y))}_G) \wedge q(x, f(x))}_F.$$

The scope of x is F , and the scope of y is G .

Notations: Quantification (cont'd)

- Given $F[x]$, a variable x is **free** if there is an occurrence of x that is not bound by any quantifier.
- **free**(F) and **bound**(F) denote the free and bound variables of F , respectively.
- It is possible that **free**(F) $\cap$ **bound**(F) $\neq \emptyset$.
 - ▶ Given $F : \forall x.p(f(x), y) \rightarrow \forall y.p(f(x), y)$, **free**(F) = $\{y\}$ and **bound**(F) = $\{x, y\}$.
- A formula F is **closed** if F has no free variables.
- Suppose **free**(F) = $\{x_1, \dots, x_n\}$. Then,
 - ▶ F 's **universal closure** is $\forall x_1 \dots \forall x_n.F$. Can be written $\forall * .F$.
 - ▶ F 's **existential closure** is $\exists x_1 \dots \exists x_n.F$. Can be written $\exists * .F$.

Interpretation

A FOL **interpretation** $I : (D_I, \alpha_I)$ is a pair of a domain D_I and an assignment α_I .

- A **domain** D_I is a nonempty set of values, such as integers or real numbers.
- An **assignment** α_I maps variables to elements of D_I . It also maps constants, function symbols, and predicate symbols to elements, functions, and predicates over D_I .
 - ▶ Each variable symbol x is assigned a value x_I from D_I .
 - ▶ Each constant is assigned a value from D_I .
 - ▶ Each n -ary function symbol f is assigned an n -ary function $f_I : D_I^n \rightarrow D_I$
 - ▶ Each n -ary predicate symbol p is assigned an n -ary predicate $p_I : D_I^n \rightarrow \{true, false\}$.

Example: Interpretation

Consider the formula

$$F : (x + y > z) \rightarrow (y > z - x)$$

that contains the binary function symbols $+$ and $-$, and the binary predicate symbol $>$, and the variables x , y , and z .

- Each symbol is just a syntactical element. Their meaning is defined by the interpretation $I = (D_I, \alpha_I)$.
- Assume the domain is the integers: $D_I = \mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$.
- Then, we may have the assignment

$$\alpha_I : \{+ \mapsto +_{\mathbb{Z}}, - \mapsto -_{\mathbb{Z}}, > \mapsto >_{\mathbb{Z}}, x \mapsto 13_{\mathbb{Z}}, y \mapsto 42_{\mathbb{Z}}, z \mapsto 1_{\mathbb{Z}}\}$$

- Semantics of FOL formulas are inductively defined as in PL.
- The cases with logical connectives ($\neg$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$) are handled in the same way as in PL.
- The semantics of predicates and quantifiers are new.

Base cases

$$I \models \top$$

$$I \not\models \perp$$

$$I \models p(t_1, \dots, t_n) \quad \text{iff} \quad \alpha_I[p(t_1, \dots, t_n)] = \text{true}$$

Inductive cases

$$I \models \neg F \quad \text{iff} \quad I \not\models F$$

$$I \models F_1 \wedge F_2 \quad \text{iff} \quad I \models F_1 \text{ and } I \models F_2$$

$$I \models F_1 \vee F_2 \quad \text{iff} \quad I \models F_1 \text{ or } I \models F_2$$

$$I \models F_1 \rightarrow F_2 \quad \text{iff} \quad I \not\models F_1 \text{ or } I \models F_2$$

$$I \models F_1 \leftrightarrow F_2 \quad \text{iff} \quad (I \models F_1 \text{ and } I \models F_2) \text{ or } (I \not\models F_1 \text{ and } I \not\models F_2)$$

$$I \models \forall x.F \quad \text{iff} \quad \text{for all } v \in D_I, I \triangleleft \{x \mapsto v\} \models F$$

$$I \models \exists x.F \quad \text{iff} \quad \text{there exists } v \in D_I \text{ such that } I \triangleleft \{x \mapsto v\} \models F$$

$$I \models p(t_1, \dots, t_n) \text{ iff } \alpha_I[p(t_1, \dots, t_n)] = \text{true}$$

- Predicates are evaluated recursively.

$$\alpha_I[p(t_1, \dots, t_n)] = \alpha_I[p](\alpha_I[t_1], \dots, \alpha_I[t_n])$$

- During evaluating terms, functions are evaluated recursively as well.

$$\alpha_I[f(t_1, \dots, t_n)] = \alpha_I[f](\alpha_I[t_1], \dots, \alpha_I[t_n])$$

$$I \models \forall x.F \text{ iff for all } v \in D_I, I \triangleleft \{x \mapsto v\} \models F$$

- $J : I \triangleleft \{x \mapsto v\}$ denotes the x -variant of I . That is, $I : (D_I, \alpha_I)$ and $J : (D_J, \alpha_J)$ agree on everything except possibly the value of the variable x . Technically,
 - ▶ $D_I = D_J$, and
 - ▶ $\alpha_I[y] = \alpha_J[y]$ for all constant, free variable, function, and predicate symbols y , except possibly x where $\alpha_J[x] = v$.
- In words, “ I is an interpretation of $\forall x.F$ iff all x -variants of I are interpretations of F ”.

$$I \models \exists x.F \text{ iff there exists } v \in D_I \text{ such that } I \triangleleft \{x \mapsto v\} \models F$$

- “ I is an interpretation of $\exists x.F$ iff some x -variant of I is an interpretation of F ”.

Example 1: Semantics

Consider the formula

$$F : (x + y > z) \rightarrow (y > z - x)$$

and the interpretation $I : (\mathbb{Z}, \alpha_I)$ where

$$\alpha_I : \{+ \mapsto +_{\mathbb{Z}}, - \mapsto -_{\mathbb{Z}}, > \mapsto >_{\mathbb{Z}}, x \mapsto 13_{\mathbb{Z}}, y \mapsto 42_{\mathbb{Z}}, z \mapsto 1_{\mathbb{Z}}\}.$$

The truth value of F under I is computed as follows:

1. $I \models y > z - x$ since $\alpha_I[y > z - x] = 42_{\mathbb{Z}} +_{\mathbb{Z}} 1_{\mathbb{Z}} >_{\mathbb{Z}} 13_{\mathbb{Z}} = \text{true}$
2. $I \models F$ by 1 and the semantics of $\rightarrow$

Example 2: Semantics

Consider the formula

$$F : \exists x. f(x) = g(x)$$

and the interpretation $I : (D : \{v_1, v_2\}, \alpha_I)$ where

$$\alpha_I : \left\{ \begin{array}{ll} f & \mapsto \{v_1 \mapsto v_1, v_2 \mapsto v_2\}, \\ g & \mapsto \{v_1 \mapsto v_2, v_2 \mapsto v_1\}, \\ = & \mapsto \{(a, b) \mapsto \text{true if } a \text{ syntactically equals } b \text{ else false}\} \end{array} \right\}$$

Compute the truth value of F under I .

Example 2: Semantics

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$$F : \exists x. f(x) = g(x)$$

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Compute the truth value of F under I .

Let J be the x -variant of I , i.e., $J : I \triangleleft \{x \mapsto v\}$ for some $v \in D$.

1. $J \not\models f(x) = g(x)$ For any $v \in D$, $\alpha_J[f(x) = g(x)] = \text{false}$
2. $I \not\models \exists x. f(x) = g(x)$ by 1 and the semantics of $\exists$

Satisfiability and Validity

- A formula F is **satisfiable** iff there exists an interpretation I such that $I \models F$.
- A formula F is **valid** iff for all interpretations I , $I \models F$.
- Technically, satisfiability and validity are defined for **closed** FOL formulas.
- But we allow two conventions for a formula F with free variables ($\text{free}(F) \neq \emptyset$).
 - ▶ If we say that a formula F is valid, we mean that its universal closure $\forall * .F$ is valid.
 - ▶ If we say that F is satisfiable, we mean that its existential closure $\exists * .F$ is satisfiable.
- Satisfiability and validity are dual as in PL.

$\forall * .F$ is valid iff $\exists * .\neg F$ is unsatisfiable

Extension of the Semantic Argument Method

Most of the proof rules from PL carry over to FOL.

$$\frac{I \models \neg F}{I \not\models F} \quad \frac{I \not\models \neg F}{I \models F} \quad \frac{I \models F \wedge G}{I \models F, I \models G} \quad \frac{I \not\models F \wedge G}{I \not\models F \mid I \not\models G}$$

$$\frac{I \models F \vee G}{I \models F \mid I \models G} \quad \frac{I \not\models F \vee G}{I \not\models F, I \not\models G} \quad \frac{I \models F \rightarrow G}{I \not\models F \mid I \models G} \quad \frac{I \not\models F \rightarrow G}{I \models F, I \not\models G}$$

$$\frac{I \models F \leftrightarrow G}{I \models F \wedge G \mid I \models \neg F \wedge \neg G} \quad \frac{I \not\models F \leftrightarrow G}{I \models F \wedge \neg G \mid I \models \neg F \wedge G}$$

$$\frac{I \models \forall x.F}{I \triangleleft \{x \mapsto v\} \models F \text{ for any } v \in D_I} \quad \frac{I \not\models \exists x.F}{I \triangleleft \{x \mapsto v\} \not\models F \text{ for any } v \in D_I}$$

$$\frac{I \models \exists x.F}{I \triangleleft \{x \mapsto v\} \models F \text{ for a fresh } v \in D_I} \quad \frac{I \not\models \forall x.F}{I \triangleleft \{x \mapsto v\} \not\models F \text{ for a fresh } v \in D_I}$$

$$\frac{J : I \triangleleft \dots \models p(s_1, \dots, s_n) \quad K : I \triangleleft \dots \not\models p(t_1, \dots, t_n)}{I \models \perp} \text{ for } i \in \{1, \dots, n\}, \alpha_J[s_i] = \alpha_K[t_i]$$

Rules for Quantifiers: “Universal” Rules

- Universal elimination I:

$$\frac{I \models \forall x.F}{I \triangleleft \{x \mapsto v\} \models F} \text{ for any } v \in D_I$$

- Existential elimination I:

$$\frac{I \not\models \exists x.F}{I \triangleleft \{x \mapsto v\} \not\models F} \text{ for any } v \in D_I$$

These rules are usually applied using a domain element v that was introduced earlier in the proof.

Rules for Quantifiers: “Existential” Rules

- Existential elimination II:

$$\frac{I \models \exists x.F}{I \triangleleft \{x \mapsto v\} \models F} \text{ for a fresh } v \in D_I$$

- Universal elimination II:

$$\frac{I \models \forall x.F}{I \triangleleft \{x \mapsto v\} \models F} \text{ for a fresh } v \in D_I$$

These rules are applied using a domain element v that has not been previously used in the proof.

- Why? Given $\exists x.F$, we choose a new value v since we do not know which value in particular satisfies F .

Contradiction Rule

$$\frac{\begin{array}{l} J : I \triangleleft \dots \models p(s_1, \dots, s_n) \\ K : I \triangleleft \dots \not\models p(t_1, \dots, t_n) \end{array}}{I \models \perp} \text{ for } i \in \{1, \dots, n\}, \alpha_J[s_i] = \alpha_K[t_i]$$

- A contradiction exists if two variants of the original interpretation I disagree on the truth value of an n -ary predicate p for a given tuple of domain values.
- A branch is **closed** if it contains a contradiction according to the contradiction rule. It is **open** otherwise.
 - ▶ In a finished proof of a valid formula, all branches must be closed.

Example 1: Semantic Argument Method

Determine the validity of the formula F .

$$F : (\forall x.p(x)) \rightarrow (\forall y.p(y))$$

Suppose F is invalid.

- | | | |
|----|--|--|
| 1. | $I \not\models F$ | assumption |
| 2. | $I \models \forall x.p(x)$ | 1 and $\rightarrow$ |
| 3. | $I \not\models \forall y.p(y)$ | 1 and $\rightarrow$ |
| 4. | $I \triangleleft \{y \mapsto v\} \not\models p(y)$ | 3 and $\forall$, for some $v \in D_I$ |
| 5. | $I \triangleleft \{x \mapsto v\} \models p(x)$ | 2 and $\forall$ |
| 6. | $I \models \perp$ | 4 and 5 |

Example 2: Semantic Argument Method

Determine the validity of the formula F .

$$F : (\forall x.p(x)) \leftrightarrow (\neg \exists x.\neg p(x))$$

Example 2: Semantic Argument Method

Determine the validity of the formula F .

$$F : (\forall x.p(x)) \leftrightarrow (\neg\exists x.\neg p(x))$$

We need to show both forward and backward directions.

$$F_1 : (\forall x.p(x)) \rightarrow (\neg\exists x.\neg p(x)), \quad F_2 : (\forall x.p(x)) \leftarrow (\neg\exists x.\neg p(x))$$

Suppose F_1 is not valid.

- | | | |
|----|---|--|
| 1. | $I \models \forall x.p(x)$ | assumption |
| 2. | $I \not\models \neg\exists x.\neg p(x)$ | assumption |
| 3. | $I \models \exists x.\neg p(x)$ | 2 and $\neg$ |
| 4. | $I \triangleleft \{x \mapsto v\} \models \neg p(x)$ | 3 and $\exists$, for some $v \in D_I$ |
| 5. | $I \triangleleft \{x \mapsto v\} \models p(x)$ | 1 and $\forall$ |
| 6. | $I \models \perp$ | 4 and 5 |

Example 2: Semantic Argument Method (cont'd)

Determine the validity of the formula F .

$$F : (\forall x.p(x)) \leftrightarrow (\neg\exists x.\neg p(x))$$

We need to show both forward and backward directions.

$$F_1 : (\forall x.p(x)) \rightarrow (\neg\exists x.\neg p(x)), \quad F_2 : (\forall x.p(x)) \leftarrow (\neg\exists x.\neg p(x))$$

Suppose F_2 is not valid.

- | | | |
|----|---|--|
| 1. | $I \not\models \forall x.p(x)$ | assumption |
| 2. | $I \models \neg\exists x.\neg p(x)$ | assumption |
| 3. | $I \triangleleft \{x \mapsto v\} \not\models p(x)$ | 1 and $\forall$, for some $v \in D_I$ |
| 4. | $I \not\models \exists x.\neg p(x)$ | 2 and $\neg$ |
| 5. | $I \triangleleft \{x \mapsto v\} \not\models \neg p(x)$ | 4 and $\exists$ |
| 6. | $I \triangleleft \{x \mapsto v\} \models p(x)$ | 5 and $\neg$ |
| 7. | $I \models \perp$ | 3 and 6 |

Example 3: Semantic Argument Method

Determine the validity of the formula F .

$$F : p(a) \rightarrow \exists x.p(x)$$

Example 3: Semantic Argument Method

Determine the validity of the formula F .

$$F : p(a) \rightarrow \exists x.p(x)$$

Assume F is invalid.

- | | | |
|----|--|---------------------|
| 1. | $I \not\models F$ | assumption |
| 2. | $I \models p(a)$ | 1 and $\rightarrow$ |
| 3. | $I \not\models \exists x.p(x)$ | 1 and $\rightarrow$ |
| 4. | $J : I \triangleleft \{x \mapsto \alpha_I[a]\} \not\models p(x)$ | 3 and $\exists$ |
| 5. | $I \models \perp$ | 2 and 4 |

Note that 2 and 4 are contradictory, since $\alpha_I[a] = \alpha_J[x]$.

Example 4: Semantic Argument Method

Show that the formula F is invalid.

$$(\forall x.p(x, x)) \rightarrow (\exists x.\forall y.p(x, y))$$

It suffices to find an interpretation I such that $I \models \neg F$. Choose $D_I = \{0, 1\}$ and $p_I = \{(0, 0), (1, 1)\}$.¹ The interpretation falsifies F .

¹We use a common notation for defining relations; $p_I(a, b)$ is *true* iff $(a, b) \in p_I$. For example, $p_I(0, 0)$ is *true* but $p_I(1, 0)$ is *false*.

Soundness and Completeness of FOL

- A proof system is **sound** if every proven formula is valid.
- A proof system is **complete** if every valid formula is provable.

Theorem (Sound)

If every branch of a semantic argument proof of $I \not\models F$ closes, then F is valid.

Theorem (Complete)

Every valid formula F has a semantic argument proof.

Substitution

- A **substitution** is a map from FOL formulas to FOL formulas.

$$\sigma : [F_1 \mapsto G_1, \dots, F_n \mapsto G_n]$$

- To compute $F\sigma$, replace each occurrence of F_i in F by G_i simultaneously.
- For example, consider the formula F and the substitution σ

$$F : (\forall x.p(x, y)) \rightarrow q(f(y), x)$$
$$\sigma : \{x \mapsto g(x), y \mapsto f(x), q(f(y), x) \mapsto \exists h(x, y)\}.$$

Then,

$$F\sigma : (\forall x.p(g(x), f(x))) \rightarrow \exists x.h(x, y)$$

Safe Substitution

- A restricted application of substitution, which has a useful semantic property.
- Idea: Before applying substitution, replace bound variables with fresh variables.
- For example, consider the formula F and the substitution σ

$$F : (\forall x.p(x, y)) \rightarrow q(f(y), x)$$
$$\sigma : \{x \mapsto g(x), y \mapsto f(x), q(f(y), x) \mapsto \exists h(x, y)\}.$$

Then, safe substitution proceeds as follows.

- 1 Renaming: $(\forall x'.p(x', y)) \rightarrow q(f(y), x)$
- 2 Substitution: $(\forall x'.p(x', f(x))) \rightarrow \exists x.h(x, y)$

Theorems for Safe Substitution

A FOL version of substitution of equivalent formulas:

Theorem

Consider the substitution

$$\sigma : \{F_1 \rightarrow G_1, \dots, F_n \rightarrow G_n\}$$

such that for each i , $F_i \iff G_i$. Then, $F \iff F\sigma$ when $F\sigma$ is computed by a safe substitution.

Theorem

If H is a valid formula schema and σ is a substitution obeying H 's side conditions, then $H\sigma$ is also valid.

- formula schema: formula templates with at least one placeholder such as $F_1, F_2, \dots$.
- side conditions: conditions specifying that certain variables do not occur free in the placeholders.

Examples: Valid Templates

- Consider the valid formula schema

$$H : (\forall x.F) \leftrightarrow (\neg \exists x. \neg F).$$

Then, the formula

$$G : (\forall x. \exists y. q(x, y)) \leftrightarrow (\neg \exists x. \neg \exists y. q(x, y))$$

is valid, because $G = H\sigma$ for $\sigma : \{F \mapsto \exists y. q(x, y)\}$.

- Consider the valid formula schema

$$H : (\forall x.F) \leftrightarrow F \text{ provided } x \notin \mathbf{free}(F).$$

Then, the formula

$$G : (\forall x. \exists y. p(z, y)) \leftrightarrow \exists y. p(z, y)$$

is valid because $G = H\sigma$ for $\sigma : \{F \mapsto \exists y. p(z, y)\}$.

Negation Normal Form (NNF)

- The normal forms of PL extend to FOL.
- A FOL formula F can be transformed into NNF by using the following equivalences.

$$\begin{array}{lll} \neg\neg F & \Longleftrightarrow & F \\ \neg\top & \Longleftrightarrow & \perp \\ \neg\perp & \Longleftrightarrow & \top \\ \neg(F_1 \wedge F_2) & \Longleftrightarrow & \neg F_1 \vee \neg F_2 \\ \neg(F_1 \vee F_2) & \Longleftrightarrow & \neg F_1 \wedge \neg F_2 \\ F_1 \rightarrow F_2 & \Longleftrightarrow & \neg F_1 \vee F_2 \\ F_1 \leftrightarrow F_2 & \Longleftrightarrow & (F_1 \rightarrow F_2) \wedge (F_2 \rightarrow F_1) \\ \neg\forall x.F[x] & \Longleftrightarrow & \exists x.\neg F[x] \\ \neg\exists x.F[x] & \Longleftrightarrow & \forall x.\neg F[x] \end{array}$$

Example: NNF

Convert the formula G into NNF where

$$G : \forall x. (\exists y. p(x, y) \wedge p(x, z)) \rightarrow \exists w. p(x, w)$$

- ① Use the equivalence $F_1 \rightarrow F_2 \iff \neg F_1 \vee F_2$.

$$\forall x. \neg (\exists y. p(x, y) \wedge p(x, z)) \vee \exists w. p(x, w)$$

- ② Use the equivalence $\neg \exists x. F[x] \iff \forall x. \neg F[x]$.

$$\forall x. (\forall y. \neg (p(x, y) \wedge p(x, z))) \vee \exists w. p(x, w)$$

- ③ Use the equivalence $\neg (F_1 \wedge F_2) \iff \neg F_1 \vee \neg F_2$.

$$\forall x. (\forall y. \neg p(x, y) \vee \neg p(x, z)) \vee \exists w. p(x, w)$$

Prenex Normal Form (PNF)

- A formula is in PNF if all of its quantifiers appear at the beginning of the formula:

$$\mathbf{Q}_1 x_1 \cdots \mathbf{Q}_n x_n . F[x_1, \cdots, x_n]$$

where $\mathbf{Q}_i \in \{\forall, \exists\}$ and F is quantifier-free.

- Every FOL F has an equivalent PNF. To convert F into PNF,
 - ① Convert F into NNF: F_1
 - ② Rename quantified variables to unique names: F_2
 - ③ Remove all quantifiers from F_2 : F_3
 - ④ Add the quantifiers in front of F_3 :

$$F_4 : \mathbf{Q}_1 x_1 \cdots \mathbf{Q}_n x_n . F_3$$

where $\mathbf{Q}_i$ are the quantifiers such that if $\mathbf{Q}_j$ is in the scope of $\mathbf{Q}_i$ in F_1 , then $i < j$.

- A FOL formula is in CNF (resp., DNF) if (1) it is in PNF and (2) its main quantifier-free subformula is in CNF (resp., DNF).

Example: PNF

Convert the formula F into PNF form.

$$F : \forall x. \neg (\exists y. p(x, y) \wedge p(x, z)) \vee \exists y. p(x, y)$$

- 1 Convert into NNF.

$$F_1 : \forall x. (\forall y. \neg p(x, y) \vee \neg p(x, z)) \vee \exists y. p(x, y)$$

- 2 Rename quantified variables.

$$F_2 : \forall x. (\forall y. \neg p(x, y) \vee \neg p(x, z)) \vee \exists w. p(x, w)$$

- 3 Remove all quantifiers.

$$F_3 : \neg p(x, y) \vee \neg p(x, z) \vee p(x, w)$$

- 4 Add the quantifiers in front of F_3 .

$$F_4 : \forall x. \forall y. \exists w. \neg p(x, y) \vee \neg p(x, z) \vee p(x, w)$$

Note that $\forall x. \exists w. \forall y. F_3$ is fine, but $\forall y. \exists w. \forall x. F_3$ is not.

Summary

FOL is an extension of propositional logic (PL) with predicates, functions, and quantifiers.

- Syntax and semantics of FOL
- Satisfiability and validity
- Substitution, Normal forms